

Alexander Totah

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Education

UC Riverside, **B.S. Electrical Engineering, Computer Science Minor**, 3.46/4.0

Sep. 2022 - May 2026

Skills

Software & Programming: C++, Python, MATLAB, Simulink, Ubuntu/Linux CLI, Arduino, SolidWorks, Docker

Circuits & Embedded Systems: Arduino, AVR Microcontrollers, Raspberry Pi, Altium, PSpice, I2C, SPI, USART

AI & Robotics: PyTorch, TensorFlow, ROS/ROS2, Nvidia Jetson Platform, Keras, OpenAI API

Technical Experience

Autonomous Vehicle Engineer, Collaborative Intelligence Systems Lab

Jan. 2024 – Present

- Coordinated development of an imitation learning pipeline using **ROS** and **Docker** with **TensorFlow** path planning algorithms, reducing lap times by 1.6 seconds over baseline.
- Optimized sensor data processing by using **Python** (PyTorch) autoencoder that compressed LiDAR data by 94% while maintaining 96% accuracy, enabling real-time operation
- Developed **PID-controlled** brushless motor system, achieving simulation-validated performance.

Systems and Control Engineer, Robotics and Medical Systems Lab

Oct. 2022 – Jan. 2024

- Designed a **PID control** system using **MATLAB** to control a steerable needle with sub-1 ms latency.
- Engineered **Arduino-based** controls for an MRI-compatible biopsy robot, reducing user error during procedures.
- Calibrated fiber optic sensor using **Simulink** to measure the bend of a steerable needle within less than 0.5mm.

Projects

Remote-Controlled Robotic Manipulator for Factory Automation

In Progress

- Designing 6-DOF robotic arm with servo actuators for material handling in manufacturing environments.
- Integrating camera feedback system for real-time monitoring of manipulation tasks.
- Developed gesture-based control system using Intel RealSense depth camera, enabling intuitive user control.

Autonomous Survey Drone, Personal Project

In Progress

- Programming flight control algorithms using **C++** while integrating GPS and camera sensors for data collection.
- Developing **OpenCV**-based computer vision pipeline for population density analysis from aerial imagery.

Just Dance, Embedded Systems Project

Project Repo Link

- Developed an interactive, motion-based game using **AVR microcontrollers** programmed in **C/C++**
- Optimized memory in LCD drawing algorithm to achieve sub-10ms refresh rate and avoid task overrun.
- Established communication protocols for accelerometer using **I2C**, display using **SPI**, and remote control using Bluetooth **USART** to reduce user latency.

Leadership Experience

Program Advisor and Lead Instructor, Kids That Code Inc.

Feb. 2024 - Present

- Advised robotics education program tailored for young learners, simplifying complex concepts and helping establish the company as a regional leader in STEM education.
- Delivered coding, robotics, and CAD instruction to 1,000+ elementary students, and coordinated assistants.

Co-Founder and Club President, Combat Robotics Club

May 2023 – May 2025

- Secured \$5,000+ in sponsorship funding from university donors, enabling expansion of robotics initiatives.
- Organized an annual competition showcasing student-created robots with 40+ spectators.